2025 WINTER SEMESTER
Programming for Data Science (BCSE207P) Lab Record
Practice Program(PP)/ Exercise Program(EP)

Date: 10/3/26 **Week No.:7** **Program S. No. 32**

Title	Student Marks Analytics Using R Data Frames
Problem Statement	You are given the marks of 10 students across four subjects. Your task is to: 1. **Create a vector for each column**, 2. **Combine all vectors into a single data frame**, and **Add computed columns**: Sum, Average, and Rank. You must use **only base R** and **data-frame operations** (no external packages, no tibbles).

Reg.No.	Name	Programme	Specialization	Subject1	Subject2	Subject3	Subject4	Subject5
24BCE1001	Aadesh Kumar	B.Tech	CSE	86	78	91	84	91
24BCE1002	Bhavya Reddy	B.Tech	AI	72	68	74	80	74
24BCE1003	Charan Iyer	B.Tech	DS	90	88	85	92	85
24BCE1004	Divya Sharma	B.Tech	IT	65	70	69	73	69
24BCE1005	Esha Nair	B.Tech	CSE	94	96	93	95	93
24BCE1006	Farhan Khan	B.Tech	ECE	58	62	60	66	60
24BCE1007	Gayathri Raj	B.Tech	Cyber	77	81	79	83	79
24BCE1008	Harish Kumar	B.Tech	AI	69	75	72	70	72
24BCE1009	Ishita Menon	B.Tech	DS	88	85	90	87	90
24BCE1010	Jai Verma	B.Tech	IT	82	79	76	85	76

Tasks

Task 1 — Create Vectors

Create one vector per column using the values from the table:

- Reg.No., Name, Programme, Specialization
- Subject1, Subject2, Subject3, Subject4
- Subject3_Marks (duplicate of Subject3, included by design)

Constraints: Use correct types (character for IDs/names/programme/specialization; numeric/integer for marks).

Task 2 — Build the Data Frame

Combine all vectors into a data frame named students.

- Ensure stringsAsFactors = FALSE.
- Print str(students) and the first 6 rows using head(students).

Task 3 — Add Computed Columns (Sum & Average)

Add a column Sum = **sum of all four subjects** (Subject1 + Subject2 + Subject3 + Subject4) for each student.

- Add a column Average = Sum / 4. Keep as numeric; you may round to 2 decimals using round() if desired.

Note: Use **vectorized operations** or rowSums(students[, c("Subject1","Subject2","Subject3","Subject4")]).
Do **not** use loops.

Task 4 — Derive Rank Based on Sum

- Add a column Rank such that the **highest Sum gets rank 1**.
- Break ties using any valid base-R approach and **state your rule** (e.g., ties.method = "min" or "dense").

Hint: students$Rank <- rank(-students$Sum, ties.method = "min")

Task 5 — Output Requirements

- Print the **complete data frame** showing columns in the order: Reg.No., Name, Programme, Specialization, Subject1, Subject2, Subject3, Subject4, Subject3_Marks, Sum, Average, Rank
- Then print the table **sorted by Rank ascending** (best student first).
  (Hint: students[order(students$Rank),])

Constraints

- **Use only data frames and base R.**
- **No external packages** (dplyr, data.table, etc.) and no file I/O.
- Keep your code **well commented**, and use **clear variable names**.

Program

```
# ---- 1) Create column vectors ----
Reg.No <-
c("24BCE1001","24BCE1002","24BCE1003","24BCE1004","24BCE10
05",

"24BCE1006","24BCE1007","24BCE1008","24BCE1009","24BCE1010
")
Name <- c("Aadesh Kumar","Bhavya Reddy","Charan Iyer","Divya
Sharma","Esha Nair",
        "Farhan Khan","Gayathri Raj","Harish Kumar","Ishita Menon","Jai
Verma")
Programme <- rep("B.Tech", 10)
Specialization <-
c("CSE","AI","DS","IT","CSE","ECE","Cyber","AI","DS","IT")

Subject1 <- c(86,72,90,65,94,58,77,69,88,82)
Subject2 <- c(78,68,88,70,96,62,81,75,85,79)
Subject3 <- c(91,74,85,69,93,60,79,72,90,76)
Subject4 <- c(84,80,92,73,95,66,83,70,87,85)
Subject3_Marks <- Subject3   # explicit column per your note

# ---- 2) Create the data frame (base R only) ----
students <- data.frame(
  Reg.No = Reg.No,
  Name = Name,
  Programme = Programme,
  Specialization = Specialization,
  Subject1 = Subject1,
  Subject2 = Subject2,
  Subject3 = Subject3,
  Subject4 = Subject4,
  Subject3_Marks = Subject3_Marks,
  stringsAsFactors = FALSE
)

str(students)
head(students)

# ---- 3) Add Sum & Average (vectorized; no loops) ----
students$Sum <- rowSums(students[,
c("Subject1","Subject2","Subject3","Subject4")])
students$Average <- students$Sum / 4  # round(students$Average, 2) if
needed

# ---- 4) Rank by Sum (highest Sum = rank 1) ----
students$Rank <- rank(-students$Sum, ties.method = "min")

# ---- 5) Print full table, then sorted by Rank ----
```

```
print(students)
students_by_rank <- students[order(students$Rank), ]
print(students_by_rank)
```

Output Snapshots

```
> # Structure of the dataframe
> str(students)
'data.frame':   10 obs. of  9 variables:
 $ Reg.No        : chr  "24BCE1001" "24BCE1002" "24BCE1003" "24BCE1004" ...
 $ Name          : chr  "Aadesh Kumar" "Bhavya Reddy" "Charan Iyer" "Divya Sharma" ...
 $ Programme     : chr  "B.Tech" "B.Tech" "B.Tech" "B.Tech" ...
 $ Specialization: chr  "CSE" "AI" "DS" "IT" ...
 $ Subject1      : num  86 72 90 65 94 58 77 69 88 82
 $ Subject2      : num  78 68 88 70 96 62 81 75 85 79
 $ Subject3      : num  91 74 85 69 93 60 79 72 90 76
 $ Subject4      : num  84 80 92 73 95 66 83 70 87 85
 $ Subject3_Marks: num  91 74 85 69 93 60 79 72 90 76
```

```
> # First 6 rows
> head(students)
     Reg.No         Name Programme Specialization Subject1 Subject2 Subject3 Subject4
1 24BCE1001 Aadesh Kumar    B.Tech            CSE       86       78       91       84
2 24BCE1002 Bhavya Reddy    B.Tech             AI       72       68       74       80
3 24BCE1003  Charan Iyer    B.Tech             DS       90       88       85       92
4 24BCE1004 Divya Sharma    B.Tech             IT       65       70       69       73
5 24BCE1005    Esha Nair    B.Tech            CSE       94       96       93       95
6 24BCE1006  Farhan Khan    B.Tech            ECE       58       62       60       66
  Subject3_Marks
1             91
2             74
3             85
4             69
5             93
6             60
```

```
> print(students)
      Reg.No         Name Programme Specialization Subject1 Subject2 Subject3 Subject4
1  24BCE1001 Aadesh Kumar    B.Tech            CSE       86       78       91       84
2  24BCE1002 Bhavya Reddy    B.Tech             AI       72       68       74       80
3  24BCE1003  Charan Iyer    B.Tech             DS       90       88       85       92
4  24BCE1004 Divya Sharma    B.Tech             IT       65       70       69       73
5  24BCE1005    Esha Nair    B.Tech            CSE       94       96       93       95
6  24BCE1006  Farhan Khan    B.Tech            ECE       58       62       60       66
7  24BCE1007 Gayathri Raj    B.Tech          Cyber       77       81       79       83
8  24BCE1008 Harish Kumar    B.Tech             AI       69       75       72       70
9  24BCE1009 Ishita Menon    B.Tech             DS       88       85       90       87
10 24BCE1010    Jai Verma    B.Tech             IT       82       79       76       85
```

	Subject3_Marks	Sum	Average	Rank
1	91	339	84.75	4
2	74	294	73.50	7
3	85	355	88.75	2
4	69	277	69.25	9
5	93	378	94.50	1
6	60	246	61.50	10
7	79	320	80.00	6
8	72	286	71.50	8
9	90	350	87.50	3
10	76	322	80.50	5

Explanation/ Observations

1. Creation of Vectors

- Separate vectors were created for each column using the `c()` function in R.
- Character vectors were used for:
  - Reg.No
  - Name
  - Programme
  - Specialization
- Numeric vectors were used for the subject marks:
  - Subject1
  - Subject2
  - Subject3
  - Subject4
- A duplicate vector **Subject3_Marks** was created from **Subject3** as required.

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2. Creation of Data Frame

- All the vectors were combined into a **data frame named** **`students`** using the `data.frame()` function.
- The argument `stringsAsFactors = FALSE` was used to ensure that text data remains as **character type** instead of factors.
- `str(students)` was used to **check the structure of the data frame**.
- `head(students)` was used to **display the first 6 rows** of the dataset.

3. Calculation of Sum and Average

- A new column **Sum** was created to calculate the **total marks of each student across the four subjects**.
- The `rowSums()` function was used to perform this calculation efficiently.
- Another column **Average** was created by **dividing the Sum by 4**, which gives the average marks of each student.

4. Ranking of Students

- The `rank()` function was used to assign ranks based on the **Sum column**.
- The negative sign `-students$Sum` ensures that **students with higher marks receive Rank 1**.
- The argument `ties.method = "min"` ensures that **students with equal total marks receive the same minimum rank**.

5. Final Output

- The final data frame contains:
  - Student details (Reg.No, Name, Programme, Specialization)
  - Marks for all subjects
  - Duplicate Subject3 column
  - Sum of marks
  - Average marks
  - Rank of each student

	• The data frame was also **sorted using** `order(students$Rank)` to display students from **highest rank to lowest rank**.
References	